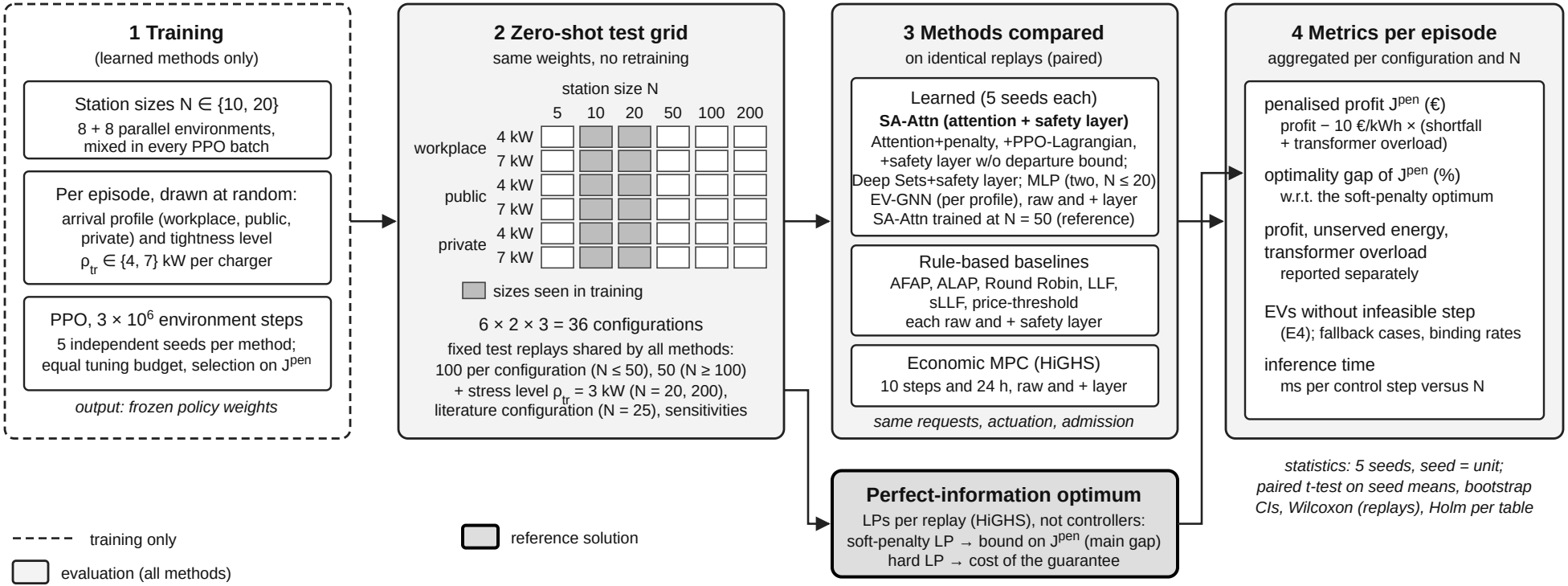
1 Training
(learned methods only)
Station sizes N ∈ {10, 20}
8 + 8 parallel environments, mixed in every PPO batch
Per episode, drawn at random:
arrival profile (workplace, public, private) and tightness level
ρ_{tr} ∈ {4, 7} kW per charger
PPO, 3 × 10^6 environment steps
5 independent seeds per method; equal tuning budget, selection on J^{pen}
output: frozen policy weights
2 Zero-shot test grid
same weights, no retraining
station size N
5 10 20 50 100 200
workplace 4 kW
7 kW
public 4 kW
7 kW
private 4 kW
7 kW
sizes seen in training
6 × 2 × 3 = 36 configurations
fixed test replays shared by all methods:
100 per configuration (N ≤ 50), 50 (N ≥ 100)
+ stress level ρ_{tr} = 3 kW (N = 20, 200),
literature configuration (N = 25), sensitivities
3 Methods compared
on identical replays (paired)
Learned (5 seeds each)
SA-Attn (attention + safety layer)
Attention+penalty, +PPO-Lagrangian,
+safety layer w/o departure bound;
Deep Sets+safety layer; MLP (two, N ≤ 20)
EV-GNN (per profile), raw and + layer
SA-Attn trained at N = 50 (reference)
Rule-based baselines
AFAP, ALAP, Round Robin, LLF,
sLLF, price-threshold
each raw and + safety layer
Economic MPC (HiGHS)
10 steps and 24 h, raw and + layer
same requests, actuation, admission
Perfect-information optimum
LPs per replay (HiGHS), not controllers:
soft-penalty LP → bound on J^{pen} (main gap)
hard LP → cost of the guarantee
4 Metrics per episode
aggregated per configuration and N
penalised profit J^{pen} (€)
profit − 10 €/kWh × (shortfall + transformer overload)
optimality gap of J^{pen} (%)
w.r.t. the soft-penalty optimum
profit, unserved energy, transformer overload
reported separately
EVs without infeasible step
(E4); fallback cases, binding rates
inference time
ms per control step versus N
statistics: 5 seeds, seed = unit;
paired t-test on seed means, bootstrap
CIs, Wilcoxon (replays), Holm per table
training only
evaluation (all methods)
reference solution